

Ejercicios de tcpdump

es una herramienta de **captura y análisis de tráfico de red** en modo texto (CLI). Sirve para **ver paquetes que pasan por una interfaz de red** en tiempo real (Ethernet/Wi-Fi), y es muy usada en administración de redes y ciberseguridad.

◆ Ejercicios básicos

1. Listar interfaces disponibles

```
tcpdump -D
```

👉 Muestra todas las interfaces de red que tcpdump puede usar.

```
jcsantos at parrot in ~
└─○ tcpdump -D
1.ens33 [Up, Running, Connected]
2.any (Pseudo-device that captures on all interfaces) [Up, Running]
3.lo [Up, Running, Loopback]
4.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
5.nflog (Linux netfilter log (NFLOG) interface) [none]
6.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
7.dbus-system (D-Bus system bus) [none]
8.dbus-session (D-Bus session bus) [none]
jcsantos at parrot in ~
```

2. Capturar tráfico en la interfaz eth0

```
tcpdump -i eth0
```

👉 Ver los primeros 20 paquetes que pasan por eth0.

```
jcsantos at parrot in ~
└─○ sudo tcpdump -i ens33
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:29:27.832416 IP 192.168.0.34.54276 > 192.168.1.255.2000: UDP, length 4
10:29:27.839619 IP 192.168.0.209.33193 > one.one.one.one.domain: 53865+ PTR? 255.1.168.192.in-addr.arpa. (44)
10:29:27.854400 IP one.one.one.one.domain > 192.168.0.209.33193: 53865 NXDomain 0/0/0 (44)
10:29:27.854591 IP 192.168.0.209.45860 > one.one.one.one.domain: 60727+ PTR? 34.0.168.192.in-addr.arpa. (43)
10:29:27.870226 IP one.one.one.one.domain > 192.168.0.209.45860: 60727 NXDomain 0/0/0 (43)
10:29:27.942239 IP 192.168.0.209.50045 > one.one.one.one.domain: 57794+ PTR? 1.1.1.1.in-addr.arpa. (38)
10:29:27.956951 IP one.one.one.one.domain > 192.168.0.209.50045: 57794 1/0/0 PTR one.one.one.one. (67)
10:29:27.957157 IP 192.168.0.209.43715 > one.one.one.one.domain: 41626+ PTR? 209.0.168.192.in-addr.arpa. (44)
10:29:27.974675 IP one.one.one.one.domain > 192.168.0.209.43715: 41626 NXDomain 0/0/0 (44)
10:29:28.306171 IP 192.168.0.209.ntp > ip212-227-145-233.pbiaas.com.ntp: NTPv4, Client, length 48
10:29:28.328766 IP ip212-227-145-233.pbiaas.com.ntp > 192.168.0.209.ntp: NTPv4, Server, length 48
^C
11 packets captured
```

3. Guardar captura en un fichero

```
tcpdump -i eth0 -w trafico.log
```

👉 Luego ábrelo con:

```
tcpdump -r trafico.log
```

```
jcsantos at parrot in ~
└─○ sudo tcpdump -i ens33 -w tcp_log.log
tcpdump: listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C13 packets captured
22 packets received by filter
0 packets dropped by kernel
jcsantos at parrot in ~
└─○ tcpdump -r tcp_log.log
reading from file tcp_log.log, link-type EN10MB (Ethernet), snapshot length 262144
10:32:12.206875 IP 192.168.0.127.netbios-ns > 192.168.1.255.netbios-ns: UDP, length 50
10:32:12.353187 ARP, Request who-has 192.168.1.163 tell 192.168.0.27, length 46
10:32:12.856312 ARP, Request who-has 192.168.1.211 tell 192.168.0.77, length 46
10:32:12.966984 IP 192.168.0.127.netbios-ns > 192.168.1.255.netbios-ns: UDP, length 50
10:32:12.983826 ARP, Request who-has 192.168.1.163 tell 192.168.0.27, length 46
10:32:13.204145 IP 192.168.0.43.44069 > 239.255.255.250.1900: UDP, length 324
10:32:13.245439 IP 192.168.0.43.54360 > 239.255.255.250.1900: UDP, length 380
10:32:13.265301 IP 192.168.0.43.54912 > 239.255.255.250.1900: UDP, length 333
10:32:13.296584 IP 192.168.0.43.37329 > 239.255.255.250.1900: UDP, length 390
10:32:13.328909 IP 192.168.0.43.51739 > 239.255.255.250.1900: UDP, length 378
10:32:13.362067 IP 192.168.0.43.44718 > 239.255.255.250.1900: UDP, length 388
10:32:13.985327 ARP, Request who-has 192.168.1.163 tell 192.168.0.27, length 46
10:32:14.626459 ARP, Request who-has 192.168.1.163 tell 192.168.0.81, length 46
jcsantos at parrot in ~
```

◆ Filtrado por protocolos y puertos

4. Capturar solo tráfico TCP en eth0

```
tcpdump -i eth0 tcp
```

```
jcsantos at parrot in ~
└─○ sudo tcpdump -i ens33 tcp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
jcsantos at parrot in ~
└─○ sudo tcpdump -i ens33 udp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:33:56.263208 IP 192.168.0.73.mdns > mdns.mcast.net.mdns: 0 [4q] PTR (QU)? _hap._tcp.local. PTR (QU)? _hap._u
dp.local. PTR (QU)? _companion-link._tcp.local. PTR (QU)? _rdlink._tcp.local. (85)
10:33:56.263796 IP6 fe80::1858:db22:e787:693a.mdns > ff02::fb.mdns: 0 [4q] PTR (QU)? _hap._tcp.local. PTR (QU)?
_hap._udp.local. PTR (QU)? _companion-link._tcp.local. PTR (QU)? _rdlink._tcp.local. (85)
10:33:56.286966 IP 192.168.0.59.mdns > mdns.mcast.net.mdns: 23 [3q] PTR (QM)? _googlecast._tcp.local. PTR (QM)?
_674A0243._sub._googlecast._tcp.local. PTR (QM)? _BE6C866D._sub._googlecast._tcp.local. (77)
10:33:56.298386 IP6 fe80::9c89:e4ff:fedd:a104.mdns > ff02::fb.mdns: 23 [3q] PTR (QM)? _googlecast._tcp.local. P
```

5. Capturar tráfico en el puerto 80 (HTTP)

```
tcpdump -i eth0 tcp port 80
```

6. Capturar tráfico DNS (UDP puerto 53)

```
tcpdump -i eth0 udp port 53
```

```
jcsantos at parrot in ~
└─$ sudo tcpdump -i ens33 udp port 50
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C
0 packets captured
0 packets received by filter
0 packets dropped by kernel
jcsantos at parrot in ~
└─$ sudo tcpdump -i ens33 tcp port 443
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:37:15.606351 IP 192.168.0.209.48466 > 191.144.160.34.bc.googleusercontent.com.https: Flags [S], seq 42840770
69, win 64240, options [mss 1460,sackOK,TS val 1881586334 ecr 0,nop,wscale 7], length 0
10:37:15.606467 IP 192.168.0.209.48478 > 191.144.160.34.bc.googleusercontent.com.https: Flags [S], seq 40167363
64, win 64240, options [mss 1460,sackOK,TS val 1881586334 ecr 0,nop,wscale 7], length 0
10:37:15.632032 IP 191.144.160.34.bc.googleusercontent.com.https > 192.168.0.209.48478: Flags [S], seq 4024823
```

◆ Filtrado por IP

7. Tráfico desde una IP concreta (host origen)

```
tcpdump -i eth0 src 192.168.43.143
```

8. Tráfico hacia una IP concreta (host destino)

```
tcpdump -i eth0 dst 192.168.43.143
```

```
jcsantos at parrot in ~
└─$ sudo tcpdump -i ens33 src 192.168.0.1
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:41:12.210216 ARP, Request who-has 192.168.0.103 tell 192.168.0.1, length 46
10:41:15.240337 ARP, Request who-has 192.168.0.103 tell 192.168.0.1, length 46
10:41:19.255775 ARP, Request who-has 192.168.0.103 tell 192.168.0.1, length 46
10:41:27.270137 ARP, Request who-has 192.168.0.103 tell 192.168.0.1, length 46
^C
4 packets captured
4 packets received by filter
0 packets dropped by kernel
jcsantos at parrot in ~
└─$ sudo tcpdump -i ens33 dst 192.168.0.1
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:42:17.024641 ARP, Request who-has 192.168.0.1 (Broadcast) tell 192.168.0.90, length 46
^C
1 packet captured
1 packet received by filter
0 packets dropped by kernel
jcsantos at parrot in ~
└─$ tcpdump -i ens33 dst 192.168.0.112
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:42:17.024641 ARP, Request who-has 192.168.0.1 (Broadcast) tell 192.168.0.90, length 46
^C
1 packet captured
1 packet received by filter
0 packets dropped by kernel
jcsantos at parrot in ~
└─$ tcpdump -i ens33 dst 192.168.0.117
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:42:17.024641 ARP, Request who-has 192.168.0.1 (Broadcast) tell 192.168.0.90, length 46
^C
1 packet captured
1 packet received by filter
0 packets dropped by kernel
jcsantos at parrot in ~
└─$
```

PORT	STATE	SERVICE	VERSION
22/tcp	filtered	ssh	

HOP	RTT	ADDRESS
1	1.02 ms	192.168.0.133

Nmap scan report for 192.168.0.219

9. Tráfico de y hacia una IP concreta (host completo)

tcpdump host 192.168.1.143

```
jcsantos at parrot in ~
└─○ sudo tcpdump host 192.168.0.1 Host is up (0.00029s latency).
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:43:51.286871 ARP, Request who-has 192.168.0.1 tell 192.168.0.72, length 46
10:43:51.315049 ARP, Request who-has 192.168.0.50 tell 192.168.0.1, length 46
10:43:52.666264 ARP, Request who-has 192.168.0.103 tell 192.168.0.1, length 46
10:43:55.317371 ARP, Request who-has 192.168.0.50 tell 192.168.0.1, length 46
10:43:55.697069 ARP, Request who-has 192.168.0.103 tell 192.168.0.1, length 46
10:43:59.723535 ARP, Request who-has 192.168.0.103 tell 192.168.0.1, length 46
10:44:03.329062 ARP, Request who-has 192.168.0.50 tell 192.168.0.1, length 46
10:44:04.136385 ARP, Request who-has 192.168.0.1 (Broadcast) tell 192.168.0.90, length 46
^C
8 packets captured
8 packets received by filter
0 packets dropped by kernel
jcsantos at parrot in ~
```

◆ Filtrado por MAC y red

10. Tráfico con destino a una dirección MAC

tcpdump ether dst 8A:B1:11:A0:BC:53

```
jcsantos at parrot in ~
└─○ sudo tcpdump ether dst 00:A0:26:D2:68:9A
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:50:10.478212 IP 192.168.0.209.42741 > one.one.one.one.domain: 1027+ Type65? csandker.io. (29)
10:50:10.478320 IP 192.168.0.209.45436 > one.one.one.one.domain: 64275+ A? csandker.io. (29)
10:50:10.478357 IP 192.168.0.209.45436 > one.one.one.one.domain: 9233+ AAAA? csandker.io. (29)
10:50:10.518826 IP 192.168.0.209.58552 > one.one.one.one.domain: 39792+ PTR? 1.1.1.1.in-addr.arpa. (38)
10:50:10.535025 IP 192.168.0.209.53084 > one.one.one.one.domain: 35427+ PTR? 209.0.168.192.in-addr.arpa. (44)
10:50:11.119744 IP 192.168.0.209.59099 > one.one.one.one.domain: 53348+ Type65? duckduckgo.com. (32)
10:50:11.119885 IP 192.168.0.209.57633 > one.one.one.one.domain: 32049+ A? duckduckgo.com. (32)
10:50:11.119947 IP 192.168.0.209.57633 > one.one.one.one.domain: 2870+ AAAA? duckduckgo.com. (32)
10:50:11.123722 IP 192.168.0.209.49424 > one.one.one.one.domain: 56159+ Type65? duckduckgo.com. (32)
10:50:11.123941 IP 192.168.0.209.58330 > one.one.one.one.domain: 43804+ A? duckduckgo.com. (32)
```

11. Tráfico de una red completa

tcpdump net 192.168.43.0/24

```
jcsantos at parrot in ~
└─○ sudo tcpdump net 192.168.0.0/24
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:52:20.673188 IP 192.168.0.98.mdns > mdns.mcast.net.mdns: 0*- [0q] 2/0/6 PTR iPad de Jon._companion-link._tcp
.local., TXT "model=J717AP" (370)
10:52:20.748682 IP 192.168.0.209.55470 > one.one.one.one.domain: 17112+ PTR? 251.0.0.224.in-addr.arpa. (42)
10:52:20.772010 IP one.one.one.one.domain > 192.168.0.209.55470: 17112 1/0/0 PTR mdns.mcast.net. (70)
10:52:20.772186 IP 192.168.0.209.49093 > one.one.one.one.domain: 62234+ PTR? 98.0.168.192.in-addr.arpa. (43)
10:52:20.796687 IP one.one.one.one.domain > 192.168.0.209.49093: 62234 NXDomain 0/0/0 (43)
10:52:20.852313 IP 192.168.0.209.38062 > one.one.one.one.domain: 51134+ PTR? 1.1.1.1.in-addr.arpa. (38)
10:52:20.875662 IP one.one.one.one.domain > 192.168.0.209.38062: 51134 1/0/0 PTR one.one.one.one. (67)
10:52:20.875893 IP 192.168.0.209.59634 > one.one.one.one.domain: 6580+ PTR? 209.0.168.192.in-addr.arpa. (44)
```


Visualización de datos

12. Mostrar el contenido en ASCII

tcpdump -i eth0 -A

```
jesantes@parrot:~$ sudo tcpdump -i ens33 -A
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:53:42.757891 IP 192.168.0.99.45134 > 192.168.0.209.8007: Flags [S], seq 2559423574, win 1024, options [mss 1460], length 0
E.....S...C...N.G...V.....
10:53:42.757911 IP 192.168.0.209.8007 > 192.168.0.99.45134: Flags [R], seq 0, ack 2559423575, win 0, length 0
F...@.@.K.....C.G.N...WP.....
10:53:42.827472 IP 0.0.0.0.bootpc > 255.255.255.255.bootps: BOOTP/DHCP, Request from 00:00:27:2d:1a:c8 (oui Unknown), length 290
E..>.@.@.8.....D.C.*mm....]B.....C.Sc5..6.....y.I()*n..9..02.
....parrot.
10:53:42.827473 IP 192.168.0.1.bootps > 255.255.255.255.bootpc: BOOTP/DHCP, Reply, length 375
E..n.....C.D.....2.....telnet-dhcp-server.....sdtp://213.0.246.180:18443x&..010002.06
.....c.Sc5..6.....3.....
10:53:42.827964 ARP, Request who-has 255.255.255.255 tell 192.168.0.90, length 46
.....3M.....Z.....
10:53:42.854987 IP 192.168.0.209.56841 > one.one.one.one.domain: 32623+ PTR? 209.0.168.192.in-addr.arpa. (44) com
E..H..@.@.V.....5.4...0.....209.0.168.192.in-addr.arpa....
10:53:42.870000 IP one.one.one.one.domain > 192.168.0.209.56841: 32623 NXDomain 0/0/0 (44) CLEAR | Secure Identity at Airports, Stadiums, & More
E..H.<@.4.....5..4.5.0.....209.0.168.192.in-addr.arpa....
10:53:42.870168 IP 192.168.0.209.40944 > one.one.one.one.domain: 33686+ PTR? 99.0.168.192.in-addr.arpa. (43)
F..G..@.@.e.....5.3.....99.0.168.192.in-addr.arpa....
10:53:42.885637 IP one.one.one.one.domain > 192.168.0.209.40944: 33686 NXDomain 0/0/0 (43) Sign In
E..G..@.@.e.....5.3.....99.0.168.192.in-addr.arpa....
10:53:42.885957 IP 192.168.0.209.43089 > one.one.one.one.domain: 2004+ PTR? 255.255.255.255.in-addr.arpa. (46) Reserve
E..J..@.@.D.....q.5.6.....255.255.255.255.in-addr.arpa....
10:53:42.892560 IP 192.168.0.99.45134 > 192.168.0.209.8652: Flags [S], seq 2559423574, win 1024, options [mss 1460], length 0
E.....g.....C...Nl...V.....
10:53:42.892599 IP 192.168.0.209.8652 > 192.168.0.99.45134: Flags [R], seq 0, ack 2559423575, win 0, length 0
E..(..@.@.K.....C..N.....WP....L.....
10:53:42.901809 IP one.one.one.one.domain > 192.168.0.209.43889: 2004 NXDomain 0/0/0 (46) TSA PreCheck
F..Ju.@.@.4..e.....5.q.6.....255.255.255.255.in-addr.arpa....
```

13. Mostrar el contenido en hexadecimal

tcpdump -i eth0 -XX

```
jesantes@parrot:~$ sudo tcpdump -i ens33 -XX
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:54:06.859381 IP 192.168.0.99.45134 > 192.168.0.209.6969: Flags [S], seq 2559423574, win 1024, options [mss 1460], length 0
0x0000: 000c 2913 85b3 0800 27d3 15dc 0800 4500 ..)...'......E.
0x0010: 002c e200 0000 2e06 2847 c0a8 0063 c0a8 ..)...(G...C..
0x0020: 00d1 b04e 1b39 988d b456 0000 0000 6002 ..)...N.9...V....
0x0030: 0400 f935 0000 0204 05b4 0000 ..)...5.....
10:54:06.859410 IP 192.168.0.209.6969 > 192.168.0.99.45134: Flags [R], seq 0, ack 2559423575, win 0, length 0
0x0000: 0800 27d3 15dc 000c 2913 85b3 0800 4500 ..)...'......E.
0x0010: 0028 0000 4000 4006 b84b c0a8 00d1 c0a8 ..)...(.@.@.K.....
0x0020: 0063 1b39 b04e 0000 0000 988d b457 5014 ..)...c.9.N.....WP.
0x0030: 0000 14df 0000 ..)...
10:54:06.941889 IP 192.168.0.209.47864 > one.one.one.one.domain: 2731+ PTR? 209.0.168.192.in-addr.arpa. (44)
0x0000: 00a0 26d2 689a 000c 2913 85b3 0800 4500 ..).h...).E.
0x0010: 0048 122d 4000 4011 64fd c0a8 00d1 0101 ..).H..@.@.d.....
0x0020: 0101 baf8 0035 0034 c3c0 0a0b 0100 0001 ..)...5.4.....
0x0030: 0000 0000 0000 0332 3039 0130 0331 3638 ..)...209.0.168
0x0040: 0331 3932 0769 6e2d 6164 6472 0461 7270 ..)...192.in-addr.ar
0x0050: 6100 000c 0001 ..)...a....
```

◆ Ejercicios combinados (operadores lógicos)

14. Tráfico desde una IP y en puerto 80

tcpdump -i eth0 src 192.168.43.143 and tcp port 443

```
jcsantos at parrot in ~ - ssh (IP-1)
└─○ sudo tcpdump -i ens33 src 192.168.0.1 and tcp port 443
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
10:56:36.948061 IP 192.168.0.1.https > 192.168.0.209.35534: Flags [R.], seq 0, ack 426953152, win 0, length 0
10:56:45.000274 IP 192.168.0.1.https > 192.168.0.209.50962: Flags [R.], seq 0, ack 3070022063, win 0, length 0
10:56:47.385046 IP 192.168.0.1.https > 192.168.0.209.50974: Flags [R.], seq 0, ack 3226842312, win 0, length 0
10:56:47.572680 IP 192.168.0.1.https > 192.168.0.209.50990: Flags [R.], seq 0, ack 4094938057, win 0, length 0
10:56:47.748772 IP 192.168.0.1.https > 192.168.0.209.50994: Flags [R.], seq 0, ack 4135475613, win 0, length 0
10:56:47.906087 IP 192.168.0.1.https > 192.168.0.209.51010: Flags [R.], seq 0, ack 1007611349, win 0, length 0
10:56:48.080226 IP 192.168.0.1.https > 192.168.0.209.51016: Flags [R.], seq 0, ack 4243774130, win 0, length 0
10:56:48.251253 IP 192.168.0.1.https > 192.168.0.209.51030: Flags [R.], seq 0, ack 954100160, win 0, length 0
10:56:48.403512 IP 192.168.0.1.https > 192.168.0.209.51042: Flags [R.], seq 0, ack 1533751273, win 0, length 0
10:56:48.570040 IP 192.168.0.1.https > 192.168.0.209.51058: Flags [R.], seq 0, ack 2374989330, win 0, length 0
```

15. Tráfico desde una IP excluyendo UDP

tcpdump -i eth0 src 192.168.1.143 and not udp

```
jcsantos at parrot in ~ - WDP RTT ADDRESS
└─○ sudo tcpdump net 192.168.0.0/24 and not udp
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:02:30.970005 ARP, Request who-has 192.168.0.46 tell 192.168.0.77, length 46
11:02:32.800379 ARP, Request who-has 255.255.255.255 tell 192.168.0.90, length 46
11:02:33.699285 IP 192.168.0.72 > 239.255.255.251: igmp v2 report 239.255.255.251
11:02:33.747925 ARP, Request who-has 192.168.1.104 tell 192.168.0.85, length 46
11:02:33.994977 ARP, Request who-has 192.168.1.176 tell 192.168.0.125, length 46
11:02:34.591757 ARP, Request who-has 192.168.1.104 tell 192.168.0.85, length 46
11:02:35.591889 ARP, Request who-has 192.168.1.104 tell 192.168.0.85, length 46
11:02:35.687078 IP 192.168.0.22 > all-routers.mcast.net: igmp leave mdns.mcast.net
11:02:35.687283 IP 192.168.0.22 > mdns.mcast.net: igmp v2 report mdns.mcast.net
11:02:36.160591 IP 192.168.0.72 > 239.255.255.250: igmp v2 report 239.255.255.250
11:02:36.182069 IP 192.168.0.72 > mdns.mcast.net: igmp v2 report mdns.mcast.net
11:02:36.571378 ARP, Request who-has 192.168.1.176 tell 192.168.0.125, length 46
11:02:37.307609 ARP, Request who-has 192.168.0.46 tell 192.168.0.77, length 46
11:02:37.498133 ARP, Request who-has 192.168.1.176 tell 192.168.0.125, length 46
11:02:37.906577 IP 192.168.0.68 > mdns.mcast.net: igmp v2 report mdns.mcast.net
^C
Filter: Hosts
15 packets captured
16 packets received by filter
0 packets dropped by kernel
jcsantos at parrot in ~
```

16. Tráfico desde una IP en HTTP o HTTPS

tcpdump -i eth0 src 192.168.1.143 and (port http or https)

```
jcsantos at parrot in ~ - WDP RTT ADDRESS
└─○ sudo tcpdump 'net 192.168.0.0/24 and (port http or https)'
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on ens33, link-type EN10MB (Ethernet), snapshot length 262144 bytes
11:05:25.005410 IP 192.168.0.209.44765 > muc12s03-in-f4.1e100.net.https: UDP, length 1252
11:05:25.005509 IP 192.168.0.209.44765 > muc12s03-in-f4.1e100.net.https: UDP, length 1252
11:05:25.005593 IP 192.168.0.209.44765 > muc12s03-in-f4.1e100.net.https: UDP, length 1252
11:05:25.005663 IP 192.168.0.209.44765 > muc12s03-in-f4.1e100.net.https: UDP, length 137
11:05:25.021116 IP muc12s03-in-f4.1e100.net.https > 192.168.0.209.44765: UDP, length 43
11:05:25.021116 IP muc12s03-in-f4.1e100.net.https > 192.168.0.209.44765: UDP, length 43
11:05:25.021116 IP muc12s03-in-f4.1e100.net.https > 192.168.0.209.44765: UDP, length 1252
11:05:25.021205 IP muc12s03-in-f4.1e100.net.https > 192.168.0.209.44765: UDP, length 1252
11:05:25.039679 IP 192.168.0.209.44765 > muc12s03-in-f4.1e100.net.https: UDP, length 1252
11:05:25.041014 IP muc12s03-in-f4.1e100.net.https > 192.168.0.209.44765: UDP, length 1252
11:05:25.041099 IP muc12s03-in-f4.1e100.net.https > 192.168.0.209.44765: UDP, length 285
11:05:25.041867 IP muc12s03-in-f4.1e100.net.https > 192.168.0.209.44765: UDP, length 617
11:05:25.041867 IP muc12s03-in-f4.1e100.net.https > 192.168.0.209.44765: UDP, length 28
11:05:25.042958 IP 192.168.0.209.44765 > muc12s03-in-f4.1e100.net.https: UDP, length 157
11:05:25.045717 IP 192.168.0.209.44765 > muc12s03-in-f4.1e100.net.https: UDP, length 695
11:05:25.056144 IP muc12s03-in-f4.1e100.net.https > 192.168.0.209.44765: UDP, length 30
```

👉 Con esta lista de **16 ejercicios** tienes un recorrido completo:

- Empezando por capturar y guardar tráfico.
- Luego filtrando por protocolo, puerto, IP, MAC y red.
- Finalmente usando operadores lógicos y visualización en distintos formatos.